**Unit:** 6 Unit 1: Ratio, Rate, Percentage **Name:** ______________
**Topic:** Rates Variation **Date:** __________

# Quiz (Habanero)

**Q1.** A glacier melts at a rate of $2.5 \cdot 10^{-3}$ meters per hour. How many meters will it melt in 12 hours?
(A) $3 \cdot 10^{-2}$ meters
(B) $3 \cdot 10^{-1}$ meters
(C) 3 meters
(D) 30 meters
(E) $3 \cdot 10^{1}$ meters

**Q2.** The temperature in a city drops by 1.8 degrees Celsius every 2 hours. What is the rate of temperature drop in degrees per hour?
(A) 0.9 degrees/hour
(B) 1.8 degrees/hour
(C) 2.5 degrees/hour
(D) 3.6 degrees/hour
(E) 4.5 degrees/hour

**Q3.** A water tank can hold 2400 liters of water. Water flows into the tank at a rate of 4 liters per minute. How many minutes will it take to fill the tank?
(A) 600 minutes
(B) 500 minutes
(C) 400 minutes
(D) 300 minutes
(E) 6000 minutes

**Q4.** The rate at which a river flows is 15 meters per second. If the river is 2.5 meters deep, what is the volume of water that flows per hour?
(A) $1.5 \cdot 10^{5}$ cubic meters/hour
(B) $1.5 \cdot 10^{6}$ cubic meters/hour
(C) $2.5 \cdot 10^{6}$ cubic meters/hour
(D) $3.5 \cdot 10^{5}$ cubic meters/hour
(E) $3.5 \cdot 10^{4}$ cubic meters/hour

**Q5.** The speed of sound in air is 343 meters per second. If a noise is heard 2.1 seconds after it is made, how far away was the source of the noise?
(A) 701.3 meters
(B) 702.3 meters
(C) 720.3 meters
(D) 721.3 meters
(E) 723.3 meters

**Q6.** A car travels 250 kilometers in 5 hours. What is its average speed in kilometers per hour?
(A) 40 km/h
(B) 45 km/h
(C) 50 km/h
(D) 55 km/h
(E) 60 km/h

**Q7.** A plane flies at a speed of 450 kilometers per hour. If it travels for 2.5 hours, what distance does it cover?
(A) 1100 kilometers
(B) 1125 kilometers
(C) 1150 kilometers
(D) 1175 kilometers
(E) 1200 kilometers

**Q8.** The population of a city grows at a rate of 2.5% per year. If the current population is 1.2 million, what will be the population in 10 years?
(A) 1.53 million
(B) 1.55 million
(C) 1.63 million
(D) 1.65 million
(E) 1.8 million

**Open Ended Questions (FRQ)**

**Q9.** A storm is causing the water level in a river to rise at a rate of 0.5 meters per hour. If the current water level is 2.5 meters, and the riverbank is 5 meters above the current water level, how many hours will it take for the water to reach the riverbank?

**Q10.** The cost of removing pollutants from a lake is directly proportional to the amount of pollutants present. If it costs

$$1200$$

to remove 150 kilograms of pollutants, how much will it cost to remove 300 kilograms of pollutants?

**Chef's Tips**

- Read the questions carefully before answering.
- Use a calculator if necessary.
- Show all your work and explanations for free response questions.